Lam Fung Academy · LF Academy

Primary 5 • 14th Lesson • Student Handout

Algebraic expression recognition + simple equations (one step)

Unit 5 • Algebra introduction • 65 minutes • 1-to-3 Online Lesson

Corresponding textbooks:"New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Volume A, Unit 5 + Modern Education, Volume 5, Unit 12

Core Traps:□ T3 Text→Algebraic Translation + T7 Reverse Thinking of Equations

SSPA Related: 🔴 High Frequency Presents the basics of algebra in the sub-test, which is required every year and accounts for about 10-15% of Paper 1

Prerequisite knowledge: Class 13 (Decimal approximation • Four arithmetic operations • Unit conversion)

Class objectives:❶ Use letters to represent unknown numbers ❷ Understand equations based on the principle of the balance ❸ Solve one-step addition, subtraction/multiplication and division equations ❹ Solve word problems using column equations

Student Name:	Class:	Date:	Time Spent:

I.Warm-Up Questions（total 5 question，5 minutes）

#	Question	Difficulty	Working Space (write out complete process)
1	Fill in the blanks with the appropriate number: (a) 8 + ___ = 15 (b) ___ − 7 = 9 (c) 20 − ___ = 13	Basic	
2	Fill in the blanks with the appropriate number: (a) 6 × ___ = 42 (b) ___ ÷ 5 = 8 (c) 72 ÷ ___ = 9	Basic	
3	"A number plus 7 equals 19" - What is this "unknown number"? How did you find out?	Basic	
4	The older brother is 3 times older than the younger brother. If the younger brother is**y**years old this year, use an algebraic expression containing**y**to express the older brother's age.	Advanced	
5	One apple sells for $a, how much does it cost to buy 5 apples? (Use algebraic expression) If a = 4, how much will be paid in total?	Advanced	

II.Core Knowledge + Worked Examples

Knowledge Point 1: Algebraic Expressions - Use letters to represent unknown numbers 🔴 SSPA

① **Why letters?**When a quantity is unknown or will change, use letters (such as x, y, a, n) to represent it

② **How to write algebraic expressions |||SEP|||: numbers × letters → numbers are written in front, and the multiplication sign is omitted. For example: 5 × y is written as**Common translations |||SEP|||: "3 more than$5y$

③ **Common translations**: "3 more than x" →$x + 3$; "2 times of y" →$2y$; "Half of n"→$\frac{n}{2}$

④ **Substitution evaluation**: When you know the numerical value of the letter, substitute it into the algebraic formula to calculate the result

□ Trap detonation example (T3: text → algebraic translation)

Write "3 times a plus 5" as an algebraic expression.

❌ Common mistakes (50% students)	✅ Correct solution
a × 3 + 5 = a8 (random merge) or 3 + 5a (order reversed) Adding a and 5 becomes a8, confusing the concept of "term"	$3a + 5$ "3 times a" = 3×a = 3a, "add 5" = +5, the result is 3a + 5 (cannot be combined!)

🧠 **Tip: "Letters represent unknown numbers, the multiplication sign is omitted and the number goes first; different letters cannot be added, so be careful when**

substituting them for evaluation."

⚠ The most frequent error: writing 3a + 5 as 8a - 3a and 5 are different categories and cannot be added!

⚠ The second most frequent error: confusing "3 times a plus 5" (3a+5) and "3 times a plus 5" (3(a+5))

Knowledge Point - Worked Examples

#	Question	Difficulty	Working Space
Example 1	Expressed algebraically: "8 is more than n" and "4 times m is 3 less".	🌱	
Example 2	If p = 6, find the value of the algebraic expression 2p + 3.	🌱	

Knowledge Point 2: Simple Equation - Balance Principle (x + a = b, x − a = b) 🔴 SSPA required test

Principle of the balance (basic properties of equations):

① Both sides of the balance are equal → Do the same operation on both sides of the equation at the same time, the equation still holds

② **Solution x + a = b**: Subtract a from both sides at the same time → x = b − a

③ **Solution x − a = b**: Add a to both sides at the same time → x = b + a

④ **Test |||SEP|||: Substitute the answer into the original equation, the left side = the right side is correct**: Substitute the answer into the original equation. The left side = the right side is correct.

①	②	③	④
write equations Translate words into equations	**reverse operation** Add change subtract change add	**Find x** Calculate the value of x	**Check calculation** Substitute into the original formula left = right?

example

Example 3: Solve the equation x + 7 = 15 (write down the steps + check)

example

Example 4: Solve the equation x − 9 = 23 (write down the steps + check)

Knowledge Point 2: Synchronization practice (must write out the step of "both sidesat the same time +/- moreless")

#	Question	Difficulty	Working Space
6	Solve the equation: x + 12 = 30	🌿	

7	Solve the equation: x − 8 = 17	🌿	
8	Solve the equation: 25 + x = 41	🌿	
9	Solve the equation: x − 15 = 9	🌿	
10	Solve the equation: x + 4.5 = 10.2	🌿	

Knowledge point 3: Simple equations—multiplication and division (ax = b, x ÷ a = b) 🔴 SSPA Advanced

Reverse thinking of multiplication and division equations (T7 trap focus):

① **Solution ax = b |||SEP|||: Divide both sides by a → x = b ÷ a**Solution x ÷ a = b |||SEP|||: Multiply both sides by a → x = b × a

② **: Many students mistakenly "divide by a" instead of "multiply by a" when x ÷ a = b**Notation

③ **: Multiply → use division to solve; division → use multiplication to solve (reverse operation!)**: Many students mistakenly "divide by a" instead of "multiply by a" when x ÷ a = b

④ **notation**: Multiplication → Use division to solve; Division → Use multiplication to solve (reverse operation!)

example

Example 5: Solve the equation 6x = 42

example

Example 6: Solve the equation x ÷ 5 = 9

T7 trap comparison questions

Solve the equation: x ÷ 3 = 12

❌ **T7 trap: reverse thinking error**

x = 12 ÷ 3 = 4

When you see "÷3", use "÷3" to solve → Big mistake! The opposite operation should be done on both sides of the equation

✅ **Correct: Reverse operation**

x = 12 × 3 = 36

x ÷ 3 = 12, ×3 on both sides: x = 12×3 = 36. Calculation: 36÷3=12 ✓

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
11	Solve the equation: 8x = 56	🌳	
12	Solve the equation: x ÷ 7 = 11	🌳	

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
13	Expressed algebraically: "x plus 15" and "7 times y".	🌱	
14	Find the values of a + 8 and 3a − 5 when a = 10.	🌱	
15	Solve the equation: x + 9 = 21	🌱	
16	Solve the equation: x − 6 = 14	🌱	
17	Solve the equation: 5x = 35	🌱	

Advanced layer (total 5 questions, 🚶🚀 choose do)

#	Question	Difficulty	Working Space
18	Solve the equation: x + 2.8 = 9.6	🌿	
19	Solve the equation: x ÷ 4 = 25	🌿	
20	Solve the equation: 4x = 100	🌿	
21	If 3n + 7 = 28, what is n? (Hint: Find the value of 3n first)	🌳	
22	Expressed algebraically: "5 times a number minus 8 equals 32." Then solve this number.	🌳	

🌳 challenge layer (total 5 questions, 🚀 choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
23	Solve the equation: 2x + 3x = 30 (hint: combine like terms first)	🌳	
24	If x ÷ 3 + 5 = 12, find x. (Hint: Two steps - first process +5, then ÷3)	🌳	
25	Solve the equation: $\frac{x}{4}$ = 7 (that is, x ÷ 4 = 7, expressed in fractional form)	🌳	
26	The sum of two consecutive numbers is 37. If the smaller number is x. (a) Express larger numbers using algebraic expressions. (b) Write an equation to find x.	🏔️	
27	The length of a rectangle is 2 times its width. If the width is w cm. (a) Express the length algebraically. (b) If the perimeter is 36 cm, use the equation to find w.	🏔️	

Knowledge point 4: Solve word problems of equations (subject to sub-test text questions) 🔴 SSPA required test

Five steps to solve the problem (T3+T7 double trap protection):

① **Let the unknown number |||SEP|||: Let x be the required quantity (start with "Let...")** Translation relationship |||SEP|||: Translate the text into algebraic expressions sentence by sentence (be careful with T3!)

② **Make equations**: Write an equation based on the equivalence relationship

③ **Solve the equation**: Use reverse operation (be careful with T7!)

④ **Check + answer sentence**: Substitute check + write "Answer:..."

⑤ **Check + Answer**: Substitute check + write "Answer:..."

example

Example 7: Xiao Ming has some stickers. After giving 15 stickers to his sister, there are still 28 stickers left. How many stickers does Xiao Ming originally have? (Suppose x → list of equations → solution → answer)

example

Example 8: A box of candies is divided equally among 6 people, and each person gets 8 pieces. How many candies are there in this box?

applicationquestionpractice (all must do, let x → column equation → solution → answer sentence)

#	Question	Difficulty	Working Space
28	Xiaomei has x yuan. After spending 35 yuan, she has 42 yuan left. How much money did she originally have?	🌿	
29	The weight of a bag of rice is n kg. 5 packages of the same rice weigh a total of 40 kg. Find n.	🌿	
30	The older brother is 3 times older than the younger brother. If the older brother is 27 years old, how old is the younger brother? (Suppose the younger brother is y years old)	🌿	
31	A rope is L cm long. After cutting off 28 cm, what remains is the original |||SEP|||. Find L. (Hint: remainder = L − 28, while remainder = $\frac{1}{3}$. Find L. (Hint: remainder = L − 28, while remainder = $\frac{L}{3}$)	🌳	
32	There are Chinese and English books on the bookshelf. There are 12 more Chinese books than English books. If there are x English books, (a) use an algebraic expression to express the number of Chinese books. (b) If there are 58 books in total, find x.	🌳	
33	A rectangle is 3 times as long as it is wide. If the perimeter is 64 cm, (a) Let the width be w and express the length algebraically. (b) Express the perimeter algebraically. (c) Use a series of equations to find w and length.	🌳	

IV. ⛰ Ultimate challenge area

#	Question	Difficulty	Working Space
⛰ 1	The sum of three consecutive numbers is 72. Let the smallest number be n. (a) Express the three numbers using algebraic expressions. (b) Make an equation to find out what each of the three numbers is.	⛰	
⛰ 2	A and B have a total of $180. The amount of A is 2 times that of B. Let B have $x. (a) List the equations. (b) Find how many dollars each A and B have.	⛰	
⛰ 3	The father is 40 years old and the son is 12 years old. How many years later will the father be twice as old as the son? (Suppose y years later → 40+y = 2(12+y))	⛰	

V. Class afterhomework

Basic must-do questions (total 5 questions, must write stepandcheck)

#	Question	Difficulty	Working Space
H1	Expressed algebraically: "m is 6 times more than 9" and "p divided by 3 plus 2".	🌱	
H2	Find the value of 5k − 8 when k = 12.	🌱	
H3	Solve the equation: x + 13 = 37	🌱	
H4	Solve the equation: x − 11 = 26	🌱	
H5	Solve the equation: 7x = 84	🌱	

Advanced choose doquestion (total 3 questions, 🚀 choose do)

#	Question	Difficulty	Working Space
H6	Solve the equations: x ÷ 3 = 18 and 4x + 3x = 49	🌿	
H7	There are x eggs in a carton. Mom used 8 and had 22 left. How many eggs are there in this box? (Solution to a column of equations)	🌳	
H8	A and B have a total of $240, and A has three times as much as B. Let B have $x. (a) List the equations. (b) Find how many dollars each person has.	🌳	

VI. The Lessoncorecommon errorsummary

✅ **Self-examination in this hall (tick after completion)**

☐ I know the pitfalls of solving each knowledge point

☐ I can complete 🌱basic questions independently

☐ I can challenge 🌿advanced questions ☐ I remember the formula

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall ☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge 🌿Advanced questions (80%+ correct) ☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	**T3: Literal → Algebraic translation error |||SEP|||: "3 times more than x 5" → 3x+5 (wrong)**: "3 times more than x 5" → 3x+5 (wrong)	"5 more than x" → x+5, "3 times of" → 3(x+5), pay attention to the brackets!
2	**Forced merging of items of different categories**：3a + 5 = 8a	3a and 5 are of different types and cannot be added! 3a+5 is the simplest
3	**T7: "Add a" when solving x+a=b**：x+7=12 → x=19	Reverse thinking: addition, change and subtraction! x+7=12 → x=12−7=5
4	**T7: Wrong division by a when solving x÷a=b**：x÷5=10 → x=2	Divide and multiply! x÷5=10 → x=10×5=50
5	**Forgot to check |||SEP|||: Don' t check after getting the x value**: Do not check after getting x value	Must be substituted into the original equation: left side = right side? If it' s not right, do it again.
6	**Algebraic expression writing format error |||SEP|||: x×7 should be written as 7x**:x×7 should be written as 7x	Numbers come first, letters come last, and the multiplication sign is omitted: 7x✓, x7 ✗
7	**The application question misses "assume x" and the answer sentence**	Must write: Let x be ___ → List equation → Solution → Answer: ______